

Lecture 04: Error and Noise

Caltech CS156 Technical Study Guide

Gemini Notebook

September 18, 2026

Executive Summary

Lecture 04 completes the formulation of supervised learning by formalizing **Error Measures** and **Noisy Targets**. It establishes that error functions are domain-specific decisions rather than purely mathematical choices (demonstrated via Supermarket vs. CIA fingerprint verification), introduces plausible vs. friendly proxy loss functions, models noisy targets using conditional probability distributions $P(y|\mathbf{x})$, and synthesizes the **Two Central Questions of Learning**.

1 Recap: The Double-Edged Sword of Transformations

Nonlinear feature transformations $\Phi : \mathcal{X} \rightarrow \mathcal{Z}$ allow linear machinery to fit arbitrary non-linear boundaries. However, transforming from d dimensions to $\tilde{d} \gg d$ dimensions increases model capacity ($\tilde{d}$ parameters).

As Yaser Abu-Mostafa noted, nonlinear transformation is like upgrading from an economy car to an 18-wheeler: it gives enormous power, but without a driver's license (statistical generalization theory), you risk catastrophic overfitting.

Table 1: Mapping Summary Between Input Space $\mathcal{X}$ and Feature Space $\mathcal{Z}$

Component	Input Space $\mathcal{X}$	Feature Space $\mathcal{Z}$
Single Point	$\mathbf{x} = (1, x_1, \dots, x_d)^T$	$\mathbf{z} = \Phi(\mathbf{x}) = (1, \phi_1(\mathbf{x}), \dots, \phi_{\tilde{d}}(\mathbf{x}))^T$
Sample Dataset	$\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N$	$\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_N$
Target Labels	$y_1, y_2, \dots, y_N$	Untouched $(y_1, y_2, \dots, y_N)$
Weight Vector	None (non-linear in $\mathbf{x}$)	$\tilde{\mathbf{w}} = (\tilde{w}_0, \tilde{w}_1, \dots, \tilde{w}_{\tilde{d}})^T$
Final Hypothesis	$g(\mathbf{x}) = \text{sign}(\tilde{\mathbf{w}}^T \Phi(\mathbf{x}))$	$\tilde{g}(\mathbf{z}) = \text{sign}(\tilde{\mathbf{w}}^T \mathbf{z})$

2 Formalization of Error Measures

An error measure quantifies how well a hypothesis h approximates the target f .

2.1 Pointwise to Overall Error

- **Pointwise Error Function:** $e(h(\mathbf{x}), f(\mathbf{x}))$ evaluates the loss on a single point $\mathbf{x}$.

- *Squared Error:* $e(h(\mathbf{x}), f(\mathbf{x})) = (h(\mathbf{x}) - f(\mathbf{x}))^2$ (used in regression).
- *Binary Error:* $e(h(\mathbf{x}), f(\mathbf{x})) = \mathbb{I}[h(\mathbf{x}) \neq f(\mathbf{x})]$ (used in classification).

- **In-Sample Error (E_{in}):** Sample average of pointwise errors over training data:

$$E_{\text{in}}(h) = \frac{1}{N} \sum_{n=1}^N e(h(\mathbf{x}_n), y_n) \quad (1)$$

- **Out-of-Sample Error (E_{out}):** Expected pointwise error over the input distribution $P(\mathbf{x})$:

$$E_{\text{out}}(h) = \mathbb{E}_{\mathbf{x} \sim P(\mathbf{x})} [e(h(\mathbf{x}), f(\mathbf{x}))] \quad (2)$$

3 Domain-Specific Error: Supermarket vs. CIA

Error measures are fundamentally **application decisions**, dictated by the user's business context. Consider fingerprint authentication ($y \in \{+1, -1\}$):

- **False Accept (Type I Error):** $f(\mathbf{x}) = -1$ (intruder), but $h(\mathbf{x}) = +1$ (accepted).
- **False Reject (Type II Error):** $f(\mathbf{x}) = +1$ (authorized user), but $h(\mathbf{x}) = -1$ (rejected).

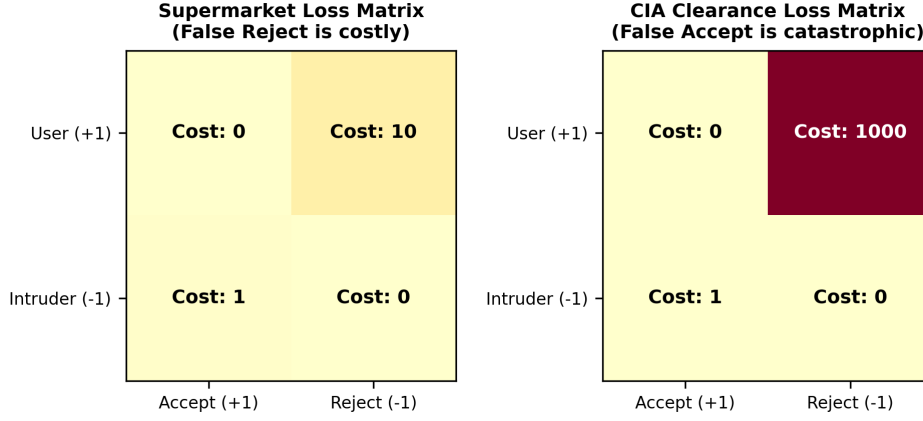


Figure 1: Comparison of Application-Specific Loss Matrices.

1. **Supermarket Loyalty Program:** A False Reject causes a paying customer immense frustration (risk of losing a \$100 customer). A False Accept merely grants an unwarranted discount. Thus, $\text{Cost}(\text{False Reject}) \gg \text{Cost}(\text{False Accept})$.
2. **CIA Security Clearance:** A False Reject merely requires an employee to re-scan their finger. A False Accept breaches national security. Thus, $\text{Cost}(\text{False Accept}) \gg \text{Cost}(\text{False Reject})$.

4 Plausible and Friendly Proxy Measures

When the exact domain cost matrix is unknown or non-differentiable, practitioners use proxy loss functions:

- **Plausible Measures:** Mathematically justified under probabilistic assumptions (e.g., Squared Error is optimal under Gaussian noise; Cross-Entropy is optimal under Bernoulli distribution).
- **Friendly Measures:** Computationally efficient measures that allow closed-form solutions (e.g., OLS) or smooth convex optimization (e.g., Logistic Regression / Log-Loss).

5 Noisy Targets & Joint Probability Distributions

In real-world problems, identical input vectors $\mathbf{x}$ can yield different outputs y due to unobserved features or measurement noise.

5.1 Target Distribution $P(y|\mathbf{x})$

Rather than a deterministic target function $y = f(\mathbf{x})$, the target is modeled as a **Conditional Target Distribution** $P(y|\mathbf{x})$.

- **Deterministic Component (Signal):** $f(\mathbf{x}) = \mathbb{E}[y|\mathbf{x}]$.
- **Stochastic Component (Noise):** $\epsilon(\mathbf{x}) = y - f(\mathbf{x})$ where $\mathbb{E}[\epsilon(\mathbf{x})] = 0$.

The data pairs $(\mathbf{x}_n, y_n)$ are sampled independently from the **Joint Probability Distribution**:

$$P(\mathbf{x}, y) = P(\mathbf{x})P(y|\mathbf{x}) \quad (3)$$

6 The Two Central Questions of Learning

With error measures and noisy targets integrated into the learning framework, machine learning decomposes into two distinct core requirements:

The Two Core Questions of Learning

1. **Is $E_{\text{out}}(g)$ close to $E_{\text{in}}(g)$?** (Theoretical Question: Generalization)
2. **Is $E_{\text{in}}(g)$ small enough?** (Practical Question: Optimization)

- **Question 1** depends on model complexity and sample size N (addressed by VC Theory).
- **Question 2** depends on the choice of learning algorithm and optimization capacity.